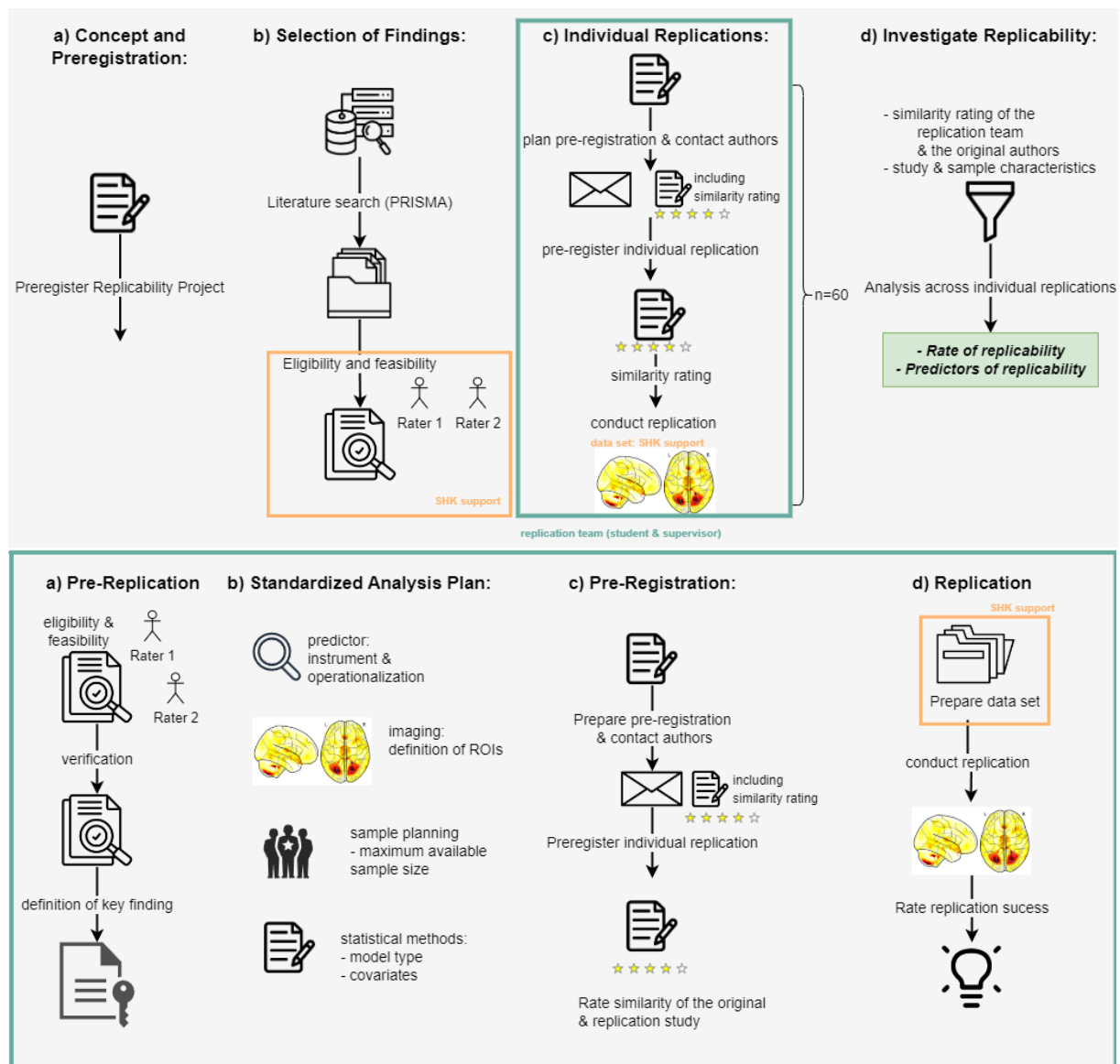


Replication-SOP



- 1) The replication team (i.e., the tandem of student and supervisor) chooses one study from the list of studies found eligible.
- 2) The analysis team has to verify whether a replication of the chosen study is possible with the available datasets (double check eligibility process of raters).
 - a. Data from five preexisting independent cohorts are available including a total of N= 6636 participants. For a detailed description of the cohorts and the data types available for the single cohorts check Sample description for the subsamples.
- 3) In case the study is verified to be feasible for replication, the replication team has to define the key finding from the original publication to be replicated. The key finding is defined as one singular statistical contrast (e.g., t-/F-contrast of one predictor or interaction X in model Y). The results from this one contrast/predictor/model may include multiple significant clusters. One cluster or multiple clusters can be defined as the key finding to be replicated. To decide which cluster or clusters to define as the key finding, the replication team should consider the size of the significant clusters, the effect size of the single clusters, and how much space the finding

takes up in the discussion section of the publication. Whether multiple clusters should be included in the definition of the key finding for replication is a case-by-case decision of the replication team. A reason to include multiple clusters might be, for example, if the clusters are located within one anatomical or functional brain region/network or are generally close to each other.

- a. The effect size (and 95% confidence interval) of the key finding has to be calculated to partial eta-squared (η_p^2) based on the reported peak voxel statistic (if multiple peak voxels are reported use the one with the largest effect size). This is done via the R package “effectsize”.
- 4) Define the methods of the replication study for the preregistration. Main criterion is the comparability to the original study (“direct replication”). Replication cohort and methods used for replication should be as similar as possible, given sufficient sample size (trade-off sample size vs. comparability). It is possible that both criteria contradict each other (e.g., larger sample available in one cohort but more comparable measure available in another cohort). In that case a judgement call will be made by the replication team (if needed after consultation of LAL¹ or JG²) weighing the advantages of the options against each other.
- a. Generally spoken, while cohort and methods should be as similar as possible, not every single characteristic of the original study can be met exactly in the replication study. Specifics will be considered if they are explicitly reported or seem of high relevance with regard to the underlying research question (e.g., distribution of gender will not be adopted exactly, but if the original sample only consists of female participants, the replication study will also only include females; the replication sample will not be matched to the original study exactly, matching all comorbidities or patterns of (number of) comorbidities, but if a study examines a cohort solely consisting of depressed patients with a comorbid social anxiety disorder, the replication cohort will be designed in the same manner).
 - b. Variables/operationalizations
 - i. Predictor: selection of instrument/measurement (e.g., CTQ)
If available, choose the same instrument as in the original study. If the same instrument is not available, choose an instrument as similar as possible. For an overview of the instruments used in the different cohorts, see Sample Description for the subsamples.
 - ii. Predictor: operationalization (e.g. CTQ-sum score)
If the same instrument is used, use the same operationalization.
 - iii. Brain variable
 1. MRI data are already preprocessed following the standard protocol currently applied at the Institute for Translational Psychiatry using the CAT12 (version: r1720) at the time of the preregistration.
 2. Regions of Interest (ROIs): Define the ROI of the main finding as similar to the original study as possible. Therefore, authors of the original study will be asked for the thresholded nifti mask of the key finding. If available, use this nifti mask for the ROI definition. If not, use the information given in the original study (peak voxels, used atlas, e.g., AAL, etc.) to define the ROIs for the replication analysis. A ROI might consist of multiple clusters within one brain region found to be significant in the original study.

¹ LAL = Luisa Altegoer

² JG = Janik Goltermann

3. Further, exploratory whole brain analyses have to be conducted.

c. Sample:

- i. Compile the replication sample aiming for a convenience sample, meaning maximum sample size available from our data, given comparable methods and sample characteristics.
- ii. Choose the cohort underlying the replication sample fitting the original sample as closely as possible. In the case that sufficient sample size (min. 90% of the original sample) cannot be reached from a single cohort, the sample can be composed from multiple cohorts. Define sample characteristics addressed in the original study (e.g., lifetime MDD vs. acute MDD, outpatient vs. inpatient treatment, MDD vs. HC, comorbidities).
- iii. If the original study features further sample specifics such as somatic inclusion/exclusion criteria these should be adopted as closely as possible.
- iv. The replication sample should be comparable regarding the age of the sample in the original study.

d. Primary statistical model

- i. Model type (regression etc.)
- ii. Definition of effect of interest (F-/t-contrast; main effect/interaction)
If the key finding is an interaction effect, clearly define whether the test statistic proofing the interaction itself or the post-hoc test further examining the direction of the effect is the key finding. Depending on this determination, the replication success criteria will be applied.
- iii. Preparation of the ROI masks for replication success testing (ROI mask of regions of interest; $r = 0$ mm, $r = 10$ mm and $r = 15$ mm sphere around the peak voxel of the original study)
→ add these as pictures to the pre-registration
- iv. Covariates (e.g., age, gender, TIV, scanner)
- v. Definition of significance: Conduct analyses at $p_{FWE} < .05$ and $p_{uncorr.} < .001$

e. Other statistical models

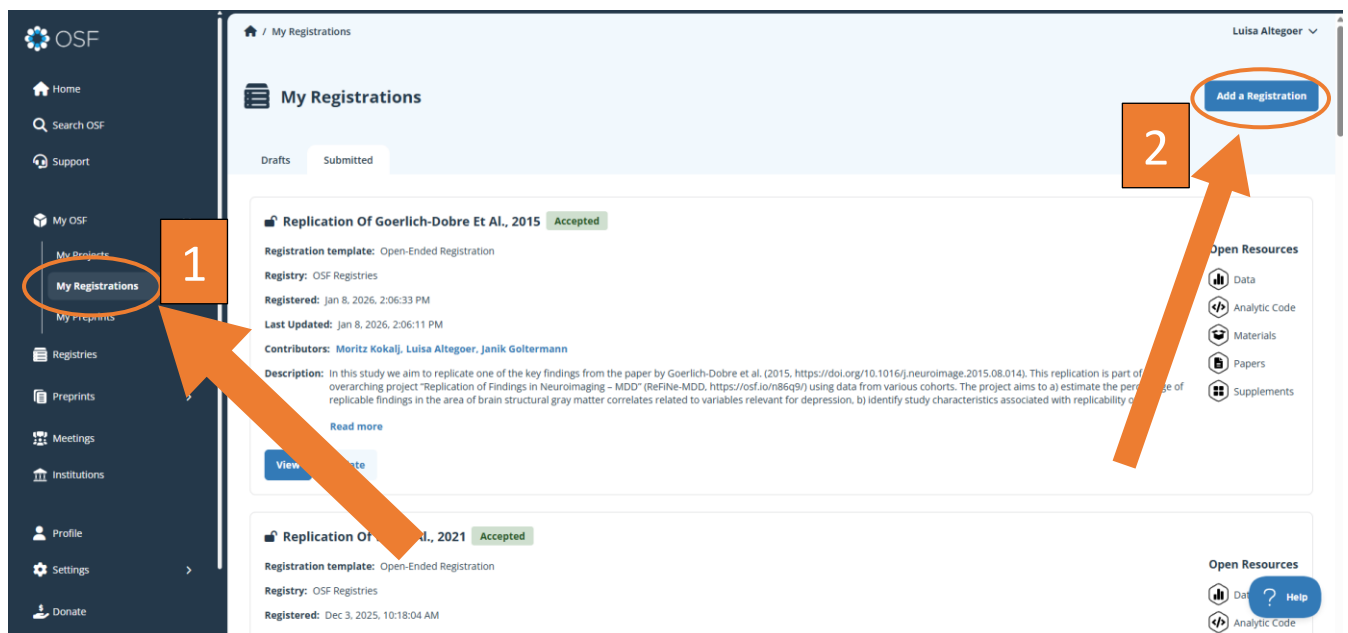
- i. Exploratory whole brain analyses
- ii. Define other statistical analyses planned beyond the primary replication analysis. These are defined at the discretion of the replication team and can include for example: other ROIs or covariates, different operationalizations of the predictor, sensitivity or subgroup analyses, other main effects or interactions (e.g., a significant effect in the ROI amygdala is defined as the key finding, but the original study also found a significant effect in two clusters in the hippocampus -> A hippocampus ROI analysis is additionally conducted). The extent of secondary analyses can be adjusted to the level of the thesis (bachelor, master or doctorate thesis).

5) Prepare the data set.

- a. Contact LAL for support with data handling via ReFiNe-MDD student workers if needed.
- b. The data set should include data from all participants included in the final sample as well as all variables needed for a sample description and for performing the specific replication analysis. The replication team can decide whether to prepare a flowchart depicting the composition of the final sample from the different cohorts.
- c. The replication team can decide whether they want to either exclude participants with missing data or use imputation algorithms. They need to consider some datasets already include imputation algorithms as part of standardized processing and quality checks (e.g. imputation algorithm for all items of questionnaires when $<10\%$ of data is missing for MACS

- data). When pooling data from different cohorts missing data and imputation should be dealt with in a consistent manner.
- d. Prepare a detailed description of sample characteristics (can be used for the thesis).
 - e. IMPORTANT: Check the predictor variable and imaging data for drop-outs (e.g., missing values, QC dropouts), but **DO NOT** conduct any analysis before the upload of the pre-registration.
- 6) Conduct preliminary power analyses using G*Power.
- a. Conduct the analysis for a power of .80 based on the sample size of the replication study.
 - b. Additionally conduct a sensitivity analysis to determine which is the smallest effect size which would be detectable given the sample size and a power of .80.
 - c. In case of a mediation analysis, use the following tool:
https://schoemanna.shinyapps.io/mc_power_med/ (source: Schoemann, A. M., Boulton, A. J., & Short, S. D. (2017). Determining Power and Sample Size for Simple and Complex Mediation Models. *Social Psychological and Personality Science*, 8(4), 379–386. <https://doi.org/10.1177/1948550617715068>)
- 7) Prepare the preregistration using the “sMRI_preregistratrion_template” (parts of this can later be directly used for the student’s thesis).
- a. Aim of the replication study
 - b. Theoretical background (short and concise)
 - c. Hypotheses and description of selected key finding
 - d. Description of sample and variables
 - e. Power analysis using g*Power (post-hoc power analysis using the effect size of the peak voxel of the original study and the sample size of the replication study)
 - f. Analysis plan
 - g. Expected replication success
- 8) Contact the authors of the original study to disclose the replication proposal.
- a. Author contacts will be coordinated by LAL. For author contact, in a first attempt the corresponding author will be contacted. If there is no response within one week, the senior author or first author will be contacted via email.
 - b. For preparation, use the template [Letter to authors](#).
 - i. First, the authors will be informed about the replication proposal. A draft of the pre-registration will be attached.
 - ii. The authors are asked to fill out a questionnaire in which they are asked to estimate the probability for replication success and the comparability/similarity between the original and replication study. The questionnaire can be accessed and submitted online.
 - c. In case that there are any missing information or open questions about the original study (e.g., sample characteristics, definition of measured constructs, analysis plan), ask the authors to provide further information. The questions will be assessed using a second online questionnaire. Hand your questions to LAL to set up the second questionnaire.
 - d. If there is no response (neither from the corresponding nor from the senior author) after three weeks, decide whether you can perform the replication without further information from the authors of the original study. If not, the replication attempt has to be discarded and excluded from the ReFiNe-MDD project (this should be the exception and only be done when there is substantial information missing within the original publication).

- 9) Preregister the replication study at OSF as a component of the ReFiNe-MDD project (<https://osf.io/n86q9/>) .
- If you do not have an account at the Open Science Framework yet, create yourself one (<https://osf.io/>).
 - Contact LAL with title and short description of the replication project, as well as names of people involved in replication (if applicable, supervisor and student). A component will be created within the overarching replication project; you and all people involved will be added as contributors.
 - Create preregistration linked to overarching replication project (LAL will create a component for each individual replication).



Add New Registration

You are submitting to OSF Registries. [Click here](#) to learn more about other hosted registries.

Step 1

Do you have content for registration in an existing OSF project?

Yes

No

Step 2

Which project do you want to register?

If your project includes components, you can select which components to include or exclude at the end of the registration.

Replication of Kuehn et al., 2012

Choose the component to be registered

Step 3

Which type of registration would you like to create?

Open-Ended Registration

Choose in the drop-down

Title

Provide a concise and descriptive title for your registration that accurately reflects the main focus of your research. A clear and specific title will help others quickly understand the essence of your study.

Replication of Kuehn et al., 2012

Description

Write a detailed description of your research project, including its purpose and any expected outcomes. This should give readers a comprehensive overview of your work.

In this study we aim to replicate one of the key findings from the paper by Kuehn et al. (2012, <https://doi.org/10.1016/j.jad.2012.03.024>) "Why ruminators won't stop: The structural and resting state correlates of rumination and its relation to depression". This replication is part of the overarching project "Replication of Findings in Neuroimaging - MDD" (ReFiNe-MDD, <https://osf.io/n86q9/>) using data from various cohorts. The project aims to a) estimate the percentage of replicable findings in brain structural gray matter correlates related to variables relevant for depression, b) identify study characteristics associated with replicability of neuroimaging findings, and c) deduct methodological guidelines to foster replicability of findings in this area of research. Therefore, a minimum of 60 single replications will be conducted. In this single replication, we will focus on the association of gray matter volume and rumination in healthy controls.

Contributors

Add all contributors (usually, student, LAL & JG)

Name	Permissions	Bibliographic Contributor	Employment history	Education history
Aleksandra Gladysz	Administrator	☒	No employment history	No education history
Luisa Altegoer	Administrator	☒	Show employment history	Show education history
Janik Goltermann	Read + Write	☒	Show employment history	Show education history

Add New Registration

1 2 3
Metadata Summary Review

Summary

Provide A Narrative Summary Of What Is Contained In This Registration Or How It Differs From Prior Registrations. If This Project Contains Documents For A Preregistration, Please Note That Here.

data from various cohorts. The project aims to a) estimate the percentage of replicable findings in brain structural gray matter correlates related to variables relevant for depression, b) identify study characteristics associated with replicability of neuroimaging findings, and c) deduct methodological guidelines to foster replicability of findings in this area of research. Therefore, a minimum of 60 single replications will be conducted. In this single replication, we will focus on the association of gray matter volume and rumination in healthy controls. Kuehn et al. (2012) found rumination to be negatively associated with clusters in bilateral inferior frontal gyrus (IFG), bilateral anterior cingulate cortex (ACC) and bilateral mid cingulate cortex (MCC) in healthy individuals. Additionally, we will examine whether this association can also be found in depressed patients.

Add Supplemental Files Or Additional Information (Optional)

Upload File
You may attach up to 5 file(s) to this question. Files cannot total over 5GB in size.
Uploaded files will automatically be archived in this registration. They will also be added to a related project that will be created for this registration.

sMRI_preregistrationKuehn_final.pdf

+ Create Folder Upload File

Almost Done...

Embargo your registration if you plan to submit to a journal that requires **blind peer review**
Do not edit any files until the registration archives.
Registrations can't be deleted, but you can withdraw.
([Learn More](#))

☒ **Make registration public immediately**
☐ **Enter registration into embargo**

Back Submit

- 10) Fill out the “pre-replication questionnaire” compiling the Similarity Rating and your expected replication success using the LimeSurvey link sent by LAL.
- 11) Perform the replication analysis following the preregistration.
- 12) Conduct a post hoc sensitivity power analysis with G*Power. Conduct the analysis for a power of .80 based on the sample size of the replication study. This will yield the effect size that can still be detected with an 80% probability with the given sample size.
 - a. In case of a mediation analysis, use the following tool:
https://schoemanna.shinyapps.io/mc_power_med/ (source: Schoemann, A. M., Boulton, A. J., & Short, S. D. (2017). Determining Power and Sample Size for Simple and Complex Mediation Models. *Social Psychological and Personality Science*, 8(4), 379–386.
<https://doi.org/10.1177/1948550617715068>)

13) Report the results of the replication using the post-replication questionnaire (via LimeSurvey, link sent via LAL).

- a. Significant effects of ROI and whole brain analyses, especially of the key finding
- b. Effect size and 95% confidence interval via R-package “effectsize” (peak voxel)
- c. To determine whether the replication was successful or not, compare the results from the original and replication study using the following criteria:
 - i. Rate whether you subjectively perceive the replication as successful.
 1. Dichotomous (yes/no)
 2. percentagewise (0-100%)
 - ii. Report if:
 1. The replication study finds a significant effect in a ROI defined as the region where the main finding was reported in the original study. If possible, this ROI should be directly defined by applying the original thresholded nifti mask provided by the original authors (orig.nii)
 2. The replication study finds a significant effect in a ROI defined by the replication team based on the information given in the original paper. If the original study finds multiple clusters within one brain region, for the replication study a ROI can be defined combining these clusters.
 3. The replication study finds a significant effect in a ROI defined as a sphere of $r = 10$ mm around the peak voxel of the original study as the center.
 4. The replication study finds a significant effect in a ROI defined as a sphere of $r = 15$ mm around the peak voxel of the original study as the center.
 5. The peak voxel found in the original study is found in the replication study as well ($p_{\text{uncorr.}} < .001$).
 - iii. All criteria except for the subjective criterion and the peak voxel will be tested on two significance level: $p_{\text{FWE}} < .05$, $p_{\text{uncorr.}} < .001$.

14) Make sure your file/data structure for the project is well organized in an established way (one folder for each model) and easily to understand for all team members.